

A Sharp Post-First-Alias Mesoscopic Crossover

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Abstract

We resolve the crossover between the first counterloop alias and the first L even coordinates immediately above it. Let $y_k = (\beta_k R)^2$, $x = (\beta R)^2 > 1$, and define

$$A_k = \frac{|s_{k,2k}|R^{2k}}{2^k}, \quad B_k(L) = \sum_{j=1}^L \frac{|s_{k,2k+2j}|R^{2k+2j}}{2^k + 2^j}.$$

The exact graded ledger gives

$$A_k = (1 - k^{-1})y_k^k, \quad \frac{B_k(L)}{A_k} = \frac{k}{k-1} \sum_{j=1}^L \frac{y_k^j}{k+j}.$$

For every integer envelope $\ell_k = o(k)$, uniformly over $1 \leq L \leq \ell_k$,

$$\frac{B_k(L)}{A_k} = \frac{x(x^L - 1)}{k(x - 1)} \{1 + o(1)\}.$$

The finite-radius factor $1 - y_k^{-L}$ is deleted only when $L \rightarrow \infty$ as well as $L = o(k)$. In that regime the sharp transition variable is $\delta_k = L - \log_x k$: the ratio tends to zero, a finite nonzero limit, or infinity according as $\delta_k \rightarrow -\infty$, $c \in \mathbb{R}$, or $+\infty$. The finite limit is $x^{c+1}/(x-1)$.

Integer depth retains a real phase. For $L_k = \lfloor \log_x k + c \rfloor$ and $\theta_k = \{\log_x k + c\}$,

$$\frac{B_k(L_k)}{A_k} = \frac{x^{c+1-\theta_k}}{x-1} \{1 + o(1)\}.$$

The phase limit set is $[0, 1]$, so the ratio has no single limit; its liminf and limsup are $x^c/(x-1)$ and $x^{c+1}/(x-1)$. On the physical noise clock the crossover lies only $2L = (2/\log x) \log \log(1/\sigma) + O(1)$ orders above the first alias. Actual-head statements remain conditional on the original unnormalized same-clock hypothesis $D_{4k}(R) \rightarrow 0$, which is not proved. Linear depth, direct/full-trace transfer, RH-288, Gates A–E, Hilbert–Polya, zero identification, and RH remain open.

1 Exact first-alias and post-alias ledgers

Fix the physical constants

$$r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad \frac{28}{17} < \lambda < \frac{17}{10}. \quad (1)$$

On the physical first-alias clock $k = k_\sigma \rightarrow \infty$, RH-17 proves

$$|M_k| = C_M \lambda^k \{1 + o(1)\}, \quad C_M > 0. \quad (2)$$

Put

$$\beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}, \quad (3)$$

and

$$x = (\beta R)^2 = \frac{(R/r_H)^2}{\lambda} > 1, \quad y_k = (\beta_k R)^2 = x \exp \left[-\frac{\log C_M}{k} + o(k^{-1}) \right]. \quad (4)$$

These are the multiplier and Hardy-normalization locks used in RH-355 [5, 7, 8].

The finite graded counterloop is

$$\mathcal{Y}_k = \{\beta_k e^{ij\pi/k}, \beta_k e^{-ij\pi/k} : 1 \leq j \leq k-1\}. \quad (5)$$

Its exact power ledger is

$$s_{k,n} = \sum_{\nu \in \mathcal{Y}_k} \nu^n = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n) \quad (6)$$

[1, 4]. Thus the first alias at $n = 2k$ has weighted modulus

$$A_k := \frac{|s_{k,2k}| R^{2k}}{2k} = \left(1 - \frac{1}{k}\right) y_k^k. \quad (7)$$

For an integer depth

$$1 \leq L \leq k-1, \quad (8)$$

define the even post-first-alias budget through $n = 2k + 2L$ by

$$B_k(L) := \sum_{j=1}^L \frac{|s_{k,2k+2j}| R^{2k+2j}}{2k+2j} = \sum_{j=1}^L \frac{y_k^{k+j}}{k+j}. \quad (9)$$

The upper limit in (8) keeps every order strictly below the second alias $4k$. Odd counterloop moments in this band vanish.

Proposition 1.1 (Exact finite ratio). *For every finite $k \geq 2$ and every depth in (8),*

$$\boxed{\frac{B_k(L)}{A_k} = \frac{k}{k-1} \sum_{j=1}^L \frac{y_k^j}{k+j}}. \quad (10)$$

Proof. In the strict upper band, (6) gives $s_{k,2k+2j} = -2\beta_k^{2k+2j}$. Hence the displayed summands in (9) are exact. Divide their sum by (7). $\square$

The common factor y_k^k cancels exactly in (10). The remaining finite-radius drift is carried by y_k^j and is negligible precisely when the post-alias depth is sublinear.

2 Uniform mesoscopic law

Theorem 2.1 (Sharp uniform post-alias law). *Let ℓ_k be any positive integer sequence with $\ell_k = o(k)$. Then*

$$\sup_{1 \leq L \leq \ell_k} \left| \frac{B_k(L)/A_k}{x(x^L - 1)/\{k(x-1)\}} - 1 \right| \rightarrow 0. \quad (11)$$

Equivalently, uniformly on that domain,

$$\boxed{\frac{B_k(L)}{A_k} = \frac{x(x^L - 1)}{k(x-1)} \{1 + o(1)\}}. \quad (12)$$

The constant C_M does not occur in the mesoscopic leading term.

Proof. Write

$$\varepsilon_k = \log(y_k/x) = -\frac{\log C_M}{k} + o(k^{-1}). \quad (13)$$

Since $\ell_k = o(k)$,

$$\sup_{1 \leq j \leq \ell_k} \left| \frac{y_k^j}{x^j} - 1 \right| = \sup_{1 \leq j \leq \ell_k} |e^{j\varepsilon_k} - 1| \longrightarrow 0. \quad (14)$$

Also

$$\sup_{1 \leq j \leq \ell_k} \left| \frac{k^2}{(k-1)(k+j)} - 1 \right| \longrightarrow 0. \quad (15)$$

All summands are positive. Therefore (10), (14), and (15) show, uniformly in L , that

$$\frac{B_k(L)}{A_k} = \frac{y_k(y_k^L - 1)}{k(y_k - 1)} \{1 + o(1)\} \quad (16)$$

$$= \frac{y_k^{L+1}}{k(y_k - 1)} (1 - y_k^{-L}) \{1 + o(1)\} \quad (17)$$

$$= \frac{1 + o(1)}{k} \sum_{j=1}^L x^j = \frac{x(x^L - 1)}{k(x - 1)} \{1 + o(1)\}. \quad (18)$$

This proves (11). The cancellation of C_M follows from (14); no numerical value of C_M is used. $\square$

The geometric subtraction in (12) must be retained at bounded depth.

Corollary 2.2 (Fixed and growing mesoscopic depths). *For every fixed integer $L \geq 1$,*

$$k \frac{B_k(L)}{A_k} \longrightarrow \frac{x(x^L - 1)}{x - 1}. \quad (19)$$

If instead

$$L \longrightarrow \infty, \quad L = o(k), \quad (20)$$

then and only at this quantified simplification step,

$$\boxed{\frac{B_k(L)}{A_k} = \frac{x}{x-1} x^{L-\log_x k} \{1 + o(1)\}.} \quad (21)$$

Proof. Equation (19) is the fixed-depth specialization of theorem 2.1. Under (20), y_k is eventually bounded below by a fixed number greater than one, so $y_k^{-L} \rightarrow 0$. Thus and only under this additional $L \rightarrow \infty$ quantifier may the factor $1 - y_k^{-L}$ in (17) be deleted. Equivalently, $x^L - 1 = x^L(1 - x^{-L}) = x^L\{1 + o(1)\}$; substitution in (12) gives (21). $\square$

In particular, deleting $1 - y_k^{-L}$ (or its limiting counterpart $1 - x^{-L}$) at fixed L would change the leading constant. No finite table or informal use of “mesoscopic” licenses that deletion.

3 The crossover and its integer phase

Theorem 3.1 (Post-first-alias crossover). *Let $1 \leq L \leq \ell_k$ for some $\ell_k = o(k)$, and put*

$$\delta_k = L - \log_x k. \quad (22)$$

Then:

1. *if $\delta_k \rightarrow -\infty$, then $B_k(L)/A_k \rightarrow 0$;*

2. if $\delta_k \rightarrow c \in \mathbb{R}$, then

$$\frac{B_k(L)}{A_k} \rightarrow \frac{x^{c+1}}{x-1}; \quad (23)$$

3. if $\delta_k \rightarrow +\infty$, then $B_k(L)/A_k \rightarrow \infty$.

The continuous balance offset is

$$L = \log_x k + \log_x \left(\frac{x-1}{x} \right) + o(1). \quad (24)$$

Proof. In the first case, (12) and $x^L - 1 \leq x^L$ give

$$0 \leq \frac{B_k(L)}{A_k} \leq \frac{x}{x-1} x^{\delta_k} \{1 + o(1)\} \rightarrow 0.$$

In the finite and supercritical cases, $L \rightarrow \infty$ automatically, so (21) applies and gives the asserted limits. Setting the finite limit equal to one yields (24). $\square$

Because the depth is integer, a fixed offset does not normally produce a single limit.

Theorem 3.2 (Floor-phase law and complete limit set). *Fix $c \in \mathbb{R}$ and define*

$$L_k(c) = \lfloor \log_x k + c \rfloor, \quad \theta_k(c) = \{\log_x k + c\}. \quad (25)$$

Then

$$\boxed{\frac{B_k(L_k(c))}{A_k} = \frac{x^{c+1-\theta_k(c)}}{x-1} \{1 + o(1)\}.} \quad (26)$$

The limit set of $\theta_k(c)$ is the closed interval $[0, 1]$. Hence

$$\liminf_{k \rightarrow \infty} \frac{B_k(L_k(c))}{A_k} = \frac{x^c}{x-1}, \quad (27)$$

$$\limsup_{k \rightarrow \infty} \frac{B_k(L_k(c))}{A_k} = \frac{x^{c+1}}{x-1}. \quad (28)$$

In particular, the full sequence has no single limit.

Proof. Here $L_k(c) \sim \log_x k$, so $L_k(c) \rightarrow \infty$ and $L_k(c) = o(k)$. Moreover $L_k(c) - \log_x k = c - \theta_k(c)$. Equation (26) follows from (21).

For $t \in (0, 1)$, take $k_m = \lfloor x^{m+t-c} \rfloor$. Then $\log_x k_m + c = m + t + o(1)$, so a subsequence of phases tends to t . For the endpoints, the choices

$$k_m^{(0)} = \lceil x^{m-c} \rceil, \quad k_m^{(1)} = \lceil x^{m+1-c} \rceil - 1 \quad (29)$$

give phases tending to zero and one, respectively. The second formula also handles the case in which x^{m+1-c} is an integer: subtracting one keeps the logarithm just below the next integer. Thus the complete phase limit set is $[0, 1]$. Apply the continuous decreasing function $t \mapsto x^{c+1-t}/(x-1)$ to obtain (27) and (28). Since $x > 1$, the endpoints differ. $\square$

On the physical clock

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1). \quad (30)$$

Therefore every bounded crossover window $L = \log_x k + O(1)$ lies at the order displacement

$$n - 2k = 2L = \frac{2}{\log x} \log \log(1/\sigma) + O(1). \quad (31)$$

This is a post-first-alias mesoscopic scale: it diverges, but is $o(k)$.

4 Conditional actual-head transfer

Let $\mathcal{H}_\sigma$ be the actual modulus-complete Hardy head of RH-342, with moments $h_{\sigma,n}$, and set

$$d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}. \quad (32)$$

The original same-clock unnormalized transport leaf is

$$D_{4k}(R) = \sum_{2 \leq n < 4k} \frac{|d_{\sigma,k,n}| R^n}{n} \longrightarrow 0 \quad (33)$$

[3, 6]. It is an open hypothesis, not a result of this paper.

Define the actual first-alias and even post-alias budgets

$$A_k^{\mathcal{H}} = \frac{|h_{\sigma,2k}| R^{2k}}{2k}, \quad (34)$$

$$B_k^{\mathcal{H}}(L) = \sum_{j=1}^L \frac{|h_{\sigma,2k+2j}| R^{2k+2j}}{2k+2j}, \quad (35)$$

and let $O_k^{\mathcal{H}}(L)$ be the corresponding weighted sum over odd orders strictly between $2k$ and $2k+2L$.

Theorem 4.1 (Conditional inheritance by the actual head). *Assume (33) on one physical clock. Then*

$$\frac{A_k^{\mathcal{H}}}{A_k} \longrightarrow 1, \quad \sup_{1 \leq L \leq k-1} \left| \frac{B_k^{\mathcal{H}}(L)}{B_k(L)} - 1 \right| \longrightarrow 0, \quad (36)$$

and

$$\sup_{1 \leq L \leq k-1} O_k^{\mathcal{H}}(L) \longrightarrow 0. \quad (37)$$

Consequently every mesoscopic conclusion of theorems 2.1, 3.1 and 3.2 transfers to $B_k^{\mathcal{H}}(L)/A_k^{\mathcal{H}}$ under this hypothesis.

Proof. The reverse triangle inequality gives

$$|A_k^{\mathcal{H}} - A_k| \leq D_{4k}(R), \quad |B_k^{\mathcal{H}}(L) - B_k(L)| \leq D_{4k}(R). \quad (38)$$

Now $A_k \sim C_M^{-1} x^k \rightarrow \infty$, while uniformly in $L \geq 1$,

$$B_k(L) \geq B_k(1) = \frac{y_k^{k+1}}{k+1} \sim \frac{x^{k+1}}{C_M k} \longrightarrow \infty.$$

Divide (38) by these lower bounds and use $D_{4k}(R) = o(1)$. Odd counterloop moments vanish, so every $O_k^{\mathcal{H}}(L)$ is bounded by $D_{4k}(R)$. Finally divide the two actual budget asymptotics to transfer the ratios. $\square$

This theorem is conditional only. It identifies neither the actual head rank nor its roots with the counterloop and does not prove (33).

5 Boundary and reproducibility

The condition $L = o(k)$ is structural. At a linear depth $L \sim \alpha k$, the two uniform replacements used in (14) and (15) fail: y_k^L/x^L can retain a $C_M^{-\alpha}$ factor, and $(k+j)^{-1}$ is not uniformly k^{-1} . No linear-depth extension is claimed. The exact ratio (10) does, however, leave a source-backed next problem: for $L/k \rightarrow \alpha \in (0, 1]$, endpoint geometric domination should be audited with the new factors $C_M^{-\alpha}$ and $(1+\alpha)^{-1}$. This separate profile is not proved here.

RH-354 controls the direct coefficient

$$p_{\sigma,k,n} = \tau_{\sigma,n} - a_n = q_{\sigma,k,n} - d_{\sigma,k,n}. \quad (39)$$

Neither the unconditional counterloop crossover nor the conditional head statement proves a theorem for p , q , or the full-trace E_{off} budget [2]. The remaining typed gluing hypotheses are still required.

The executable artifact checks the exact rational ledger at $\lambda = 5/3$, $C_M = 1$, the full geometric factor $x^L - 1$, and synthetic finite-radius phase rows at archived high-precision diagnostics. The finite ratio itself is not required to lie inside its limiting phase cluster at any prescribed k ; only convergence to the phase law is tested. The decimals are not interval certificates, actual noisy-head observations, or asymptotic evidence.

RH-356 proves an unconditional mesoscopic crossover only for the graded counterloop and a conditional transfer statement for the actual head. It does not prove $D_{4k}(R) \rightarrow 0$, root or rank matching, a linear-depth law, direct/full-trace closure, the RH-241 moving noisy envelope, or RH-288 activation. Gates A–E remain false/open. No Hilbert–Polya operator, Riemann-zero identification, von Mangoldt trace, completed-zeta divisor equality, or proof of the Riemann Hypothesis is claimed.

References

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